F78AA - Actuarial and Financial Mathematics A
Tutorial 7 – IRR, DPP and profits

Question 1 You invest £1 000 in a project and receive back £218.00 at the end of the first year of the project, £356.43 at the end of the second year and £647.51 at the end of the third year.

1. Write down an equation for the net present value (NPV) from the project and evaluate it at an interest rate of: (a) 8% p.a. effective; and (b) 9% p.a. effective.
2. Calculate the internal rate of return (IRR) on the project.

Question 2 You borrow £1 000 from your bank at an effective rate of interest of 6% p.a. Using your borrowings, you invest in a project which pays £2 158.92 at the end of ten years.

1. Calculate your internal rate of return (IRR).

You repay the bank at the end of ten years (repaying the loan and the interest accumulated over the ten years) out of the payment from the project.

2. Calculate your profit at the end of the ten years.

Question 3 You borrow £1 000 from the bank at an effective rate of 6% p.a. and invest it in a project. The project provides you with a payment of £400 at the end of the first year. From this payment you repay the interest you are owing to the bank and use the balance of the payment to repay as much as you can of the loan from the bank. At the end of the second year you receive £800 from the project. From this payment, you repay both the interest you are owing to the bank and the outstanding loan to the bank. What you have left over, i.e. the balance, is profit for you.

1. Calculate the profit.
2. Calculate your internal rate of return (IRR) on the project.

Question 4 You borrow £1 000 000 for a company that will make profits of £200 000 at the end of each of the next 10 years. The lender charges interest at 7% per annum effective. You will use the profits to repay the loan as soon as possible, and then any further profits can be invested to earn interest at 6% per annum effective.

1. Calculate the Discounted Payback Period (DPP).
2. Calculate the accumulated profit at the end of 10 years.

Question 5 You borrow £100 million to set up a factory which will make profits of £15 million at the end of each of the next 10 years. You agree to repay the loan by annual level repayments over the 10 years of the venture, calculated at an interest rate of 5% per annum effective. The net profits are invested to earn 3% per annum effective until the end of 10 years.

1. Calculate your accumulated profit at the end of 10 years.
2. Calculate how much more accumulated profit you would have made if you had used all of the profits in the early years to repay the loan as fast as possible.

Question 6 You wish to borrow £10 million for a project, which will return profits of £1.5 million at the end of each of the first 5 years, and £2 million at the end of each of the next five years. You want to maximise your accumulated profits at the end of the project.

You have three possible sources of loan capital.

(I) Borrow it from bank A, which charges interest at 8% per annum, and use the net profits from the venture to repay the loan as soon as possible.

(II) Borrow it from bank B, which charges interest at 7.5% per annum, and repay it by level annual repayments over the ten years.

(III) Issue a bond with term 10 years and a coupon of 7% per annum, payable annually. The bond will be issued at par and redeemed at 105%.

Net profits can be invested to accumulate at 5% per annum effective. What method of financing should you choose?

Question 7 You borrow £100 million and build a widget factory. It will return profits at the end of each of the next 10 years, starting with £14 million in the first year and subsequently increasing by 5% compound each year.

You borrow the money from a bank which charges interest of 7% per annum on loans, and you will use the profits to repay the loan as quickly as possible. Further profits can then be invested at 5% per annum effective.

Calculate your accumulated profit at the end of 10 years.

Solutions: Q1.1(a). 21.447; Q1.1(b). -0.003; Q1.2. 9%; Q2.1. 8.000%; Q2.2. £368.07; Q3.1. £100.40; Q3.2. 11.652%; Q4.1. 7 years; Q4.2. £785 623; Q5.1. £23.50 million; Q5.2. £2.073 million; Q6. II; Q7. £39.721 million.

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Solutions to Tutorial 7 – IRR, DPP and profits

Solution 1

1. At an effective rate of interest i, the equation for the NPV is (in £)

$$\text{NPV}(i) := -1\,000 + 218v + 356.43v^2 + 647.51v^3, \qquad \text{where } v = (1+i)^{-1}.$$

   (a) NPV(0.08) = 21.447; and (b) NPV(0.09) = −0.003.

2. Part (b) above shows that the IRR is 9% per annum effective, without the need for any more calculations.

Solution 2

1. The IRR is the solution of NPV$(i) = 0$, where NPV$(i) := -1\,000 + 2\,158.92v^{10}$. Solving,

$$v^{10} = 1\,000/2\,158.92 \qquad \Rightarrow \qquad v = 0.92593 \qquad \Rightarrow \qquad \text{IRR} = 1/0.92593 - 1 = 0.08000$$

2. You repay to the bank the amount $£1\,000 \times 1.06^{10} = 1\,790.85$ so the profit is $2\,158.92 - 1\,790.85 = £368.07$.

Solution 3

1. Time is measured in years. At time 1 the interest owed to the bank is £60. You pay this from the £400 received, leaving £340 in your pocket. Using the leftover £340, you pay back part of the loan. The value of the outstanding loan at time 1 is then $1\,000 - 340 = £660$.

   At time 2, the interest on the outstanding loan is £39.60, and the total value of the loan and interest is thus £699.60. You pay this off from the £800 received, leaving £100.40 profit in your pocket.

2. The IRR is the solution of NPV$(i) = 0$, where NPV$(i) := -1\,000 + 400v + 800v^2$.

   As this is a quadratic equation, we solve it exactly. We solve it to find that the only positive root is $v = 0.89564$. Hence $i = 0.11652$.

Solution 4 Time is measured in years and we work in £000s.

1. As payments are only received at integer times, then the DPP must also be at an integer time. Up to the DPP, we use an effective interest rate of 7% p.a.

   Begin by finding the range of values of time t such that the accumulated value of the profits received at time t exceeds the accumulated borrowings at time t, i.e.

$$200\, s_{\overline{t}|}^{0.07} \geq 1\,000 \times 1.07^t.$$

   This can be equivalently written as the values at time 0, i.e.

$$200\, a_{\overline{t}|}^{0.07} \geq 1\,000 \qquad \Rightarrow \qquad 200\, \frac{1 - 1.07^{-t}}{0.07} \geq 1\,000.$$

We solve for the range of values of t, so we want to isolate t on one side of the inequality.

NOTE Be careful when dividing or multiplying across an inequality by a negative number, as it reverses the inequality. For example, what values does $\ln(x)$ take, for $x < 1$? For $x = 1$? For $x > 1$?

More generally, you have to check if the inequality is reversed if you apply a function to both sides of the inequality. Beware!

Back to the problem:

$$1 - 1.07^{-t} \geq \frac{0.07 \times 1\,000}{200} \qquad \Rightarrow \qquad 0.65 \geq 1.07^{-t}.$$

Taking natural logs of both sides (which does not reverse the inequality as the log function is a monotonically increasing function, and hence preserves the ordering of the inequality),

$$\ln(0.65) \geq -t \times \ln(1.07) \qquad \Rightarrow \qquad -t \leq \frac{\ln(0.65)}{\ln(1.07)} \qquad \Rightarrow \qquad t \geq -\frac{\ln(0.65)}{\ln(1.07)} = 6.367.$$

As we need a time at which payments are made, i.e. an integer time, the DPP is the first integer *after* 6.367 (after, since $t \geq 6.367$), i.e. DPP= 7 years. Don't forget the unit of time!

NOTE If you get a negative DPP, then something has gone wrong in your calculation.

2. The profit remaining after the loan has been repaid is

$$200\, s_{\overline{7}|}^{0.07} - 1\,000 \times 1.07^7 = 125.022,$$

so the accumulated profit after 10 years is (now using the 6% effective interest rate as the loan has been repaid)

$$125.022 \times 1.06^3 + 200 s_{\overline{3}|}^{0.06} = £785.623.$$

Solution 5 Time is measured in years and we work in £millions.

1. The annual repayment is $100/a_{\overline{10}|}^{0.05} = 12.950$. At the end of each year, you receive £15, but make the repayment of 12.950. Therefore, there is £2.050 left over each year. Thus £2.050 is your annual net profit. The net profit is invested at 3% per annum, with the 10 years' worth of net profit accumulating to

$$2.050\, s_{\overline{10}|}^{0.03} = £23.501 \text{ million}.$$

2. The loan would have been repaid at an unknown time t, which satisfies

$$100 \times 1.05^t \leq 15\, s_{\overline{t}|}^{0.05} \quad \text{or, equivalently,} \quad 100 \leq 15\, a_{\overline{t}|}^{0.05}.$$

Expanding the annuity symbol and isolating t on one side of the inequality,

$$1 - 1.05^{-t} \geq \frac{0.05 \times 100}{15} \qquad \Rightarrow \qquad \frac{2}{3} \geq 1.05^{-t}.$$

Taking (natural) logs of both sides of the last inequality,

$$\ln(2/3) \geq -t \times \ln(1.05) \qquad \Rightarrow \qquad -t \leq \frac{\ln(2/3)}{\ln(1.05)} \qquad \Rightarrow \qquad t \geq 8.310.$$

As we need a time at which payments are made, i.e. here at integer times, the DPP is 9 years.

The profit remaining after the loan has been repaid is

$$15\, s_{\overline{9}|}^{0.05} - 100 \times 1.05^9 = 10.266$$

so the accumulated profit after 10 years is $10.266 \times 1.03 + 15 = £25.574$.

Therefore, you would have an extra accumulated profit of about $25.574 - 23.501 = £2.073$ million if you repay the loan as soon as possible.

Solution 6 Time is measured in years and we work in £millions. We must work out the accumulated profit under each financing scheme.

(I) Here, we work out the DPP in order to calculate the profit. First find the capital outstanding at time 5:

$$10 \times 1.08^5 - 1.5 s_{\overline{5}|}^{0.08} = 5.893.$$

As this is positive, then the loan will be repaid past time 5. Thus the loan will be repaid at time $5 + t$, which satisfies

$$5.893 \times 1.08^t \leq 2\, s_{\overline{t}|}^{0.08} \quad \text{or, equivalently,} \quad 5.893 \leq 2\, a_{\overline{t}|}^{0.08}.$$

Expanding the annuity symbol and isolating t on one side of the inequality,

$$1 - 1.08^{-t} \geq \frac{0.08 \times 5.893}{2} \qquad \Rightarrow \qquad 0.76428 \geq 1.08^{-t}.$$

Taking (natural) logs of both sides of the last inequality,

$$\ln(0.76428) \geq -t \times \ln(1.08) \qquad \Rightarrow \qquad -t \leq \frac{\ln(0.76428)}{\ln(1.08)} \qquad \Rightarrow \qquad t \geq 3.493.$$

As we need a time at which payments are made, i.e. an integer time in this example, the loan is repaid 4 years after time 5, i.e. at time 9 years.

Doing the calculations starting from time 5, the profit remaining after the loan has been repaid is

$$2\, s_{\overline{4}|}^{0.08} - 5.893 \times 1.08^4 = 0.995,$$

so the accumulated profit after 10 years is $0.995 \times 1.05 + 2 = £3.045$ million.

(II) The annual repayment is $10/a_{\overline{10}|}^{0.075} = 1.457$. Therefore the accumulated profit will be

$$(1.5-1.457)\, s_{\overline{5}|}^{0.05} \times 1.05^5 + (2-1.457)\, s_{\overline{5}|}^{0.05} = \big(0.043 \times 1.05^5 + 0.543\big) \times s_{\overline{5}|}^{0.05} = £3.304 \text{ million.}$$

(III) Issuing £10 nominal of the bond, The accumulated profit after 10 years is:

$$(1.5-0.7)\, s_{\overline{5}|}^{0.05} \times 1.05^5 + (2-0.7)\, s_{\overline{5}|}^{0.05} - 10 \times 1.05 = \big(0.8 \times 1.05^5 + 1.3\big)\, s_{\overline{5}|}^{0.05} - 10.5 = £2.325 \text{ million.}$$

Therefore you should choose financing option (II), as it results in the highest profit.

Solution 7 Work in £millions. The capital outstanding just after the tth year's profit has been used to repay the loan is

$$100(1.07)^t - 14(1.07)^{t-1} - 14(1.05)(1.07)^{t-2} - \cdots - 14(1.05)^{t-1}.$$

The loan will be repaid when the capital outstanding first falls below zero, i.e. the first time t at which

$$100(1.07)^t - 14(1.07)^{t-1} - 14(1.05)(1.07)^{t-2} - \cdots - 14(1.05)^{t-2}(1.07)^1 - 14(1.05)^{t-1} \leq 0.$$

As before, we want to find the range of t for which the inequality holds. We look for a geometric progression. Dividing both sides of the inequality by $(1.07)^t$ and multiply by 1.05, we get

$$100\times 1.05 \leq 14(1.05)(1.07)^{-1}+14(1.05)^2(1.07)^{-2}+\cdots+14(1.05)^{t-1}(1.07)^{-t+1}+14(1.05)^t(1.07)^{-t},$$

i.e.

$$105 \leq 14\left(\frac{1.05}{1.07} + \frac{1.05^2}{1.07^2} + \cdots + \frac{1.05^{t-1}}{1.07^{t-1}} + \frac{1.05^t}{1.07^t}\right).$$

Setting $j := 1.07/1.05 - 1 = 0.019048$, the inequality can be written

$$105 \leq 14\left(\frac{1}{1+j} + \frac{1}{(1+j)^2} + \cdots + \frac{1}{(1+j)^{t-1}} + \frac{1}{(1+j)^t}\right) = 14\, a^j_{\overline{t}|}.$$

i.e. $105 \leq 14\, a^j_{\overline{t}|}$. Isolating t,

$$\begin{aligned} \frac{105 \times 0.019048}{14} \leq 1 - (1.019048)^{-t} \quad &\Rightarrow \quad (1.019048)^{-t} \leq 0.85714 \\ &\Rightarrow \quad t \geq -\frac{\ln(0.85714)}{\ln(1.019048)} = 8.170. \end{aligned}$$

As we need the DPP to be at a time at which payments are made, the loan is repaid at time 9 years. The profit left after repaying the loan at time 9 is

$$\begin{aligned} & -100(1.07)^9 + 14(1.07)^8 + 14(1.05)(1.07)^7 + \cdots + 14(1.05)^8 \\ = \; & -100(1.07)^9 + 14(1.07)^9\left((1.07)^{-1} + (1.05)(1.07)^{-2} + \cdots + (1.05)^8(1.07)^{-9}\right) \\ = \; & -100(1.07)^9 + \frac{14(1.07)^9}{1.05}\left((1.05)(1.07)^{-1} + (1.05)^2(1.07)^{-2} + \cdots + (1.05)^9(1.07)^{-9}\right) \\ = \; & -100(1.07)^9 + \frac{14(1.07)^9}{1.05} a^{0.019048}_{\overline{9}|} \\ = \; & 17.145 \text{ million.} \end{aligned}$$

Therefore the accumulated profit at the end of 10 years is

$$17.145 \times 1.05 + 14\,(1.05)^9 = £39.721 \text{ million.}$$